

FORENSIC BIAS AUDIT — STAKEHOLDER DEFENSE

COMPAS-Like Pretrial Risk Assessment Model

Modeled on the system California's DTC confirmed is in active CDCR use — June 2026 (Johnson, 2026)

RECOMMENDATION

CONDITIONAL GO

Deploy under Equal Opportunity thresholds, with mandatory NIST AI RMF validation, audit, and human-oversight controls in force before go-live.

85%

reduction in cross-group
error disparity (D_fair)

0.63 pt

accuracy cost of the
fairness correction

3

NIST AI RMF governance
controls required pre-launch

What This System Does

- **Purpose**

Estimates the likelihood a defendant will reoffend, to inform pretrial release and bail decisions.

- **Inputs**

Criminal-history features (e.g., prior convictions, juvenile record) alongside demographic fields including race, age, and sex.

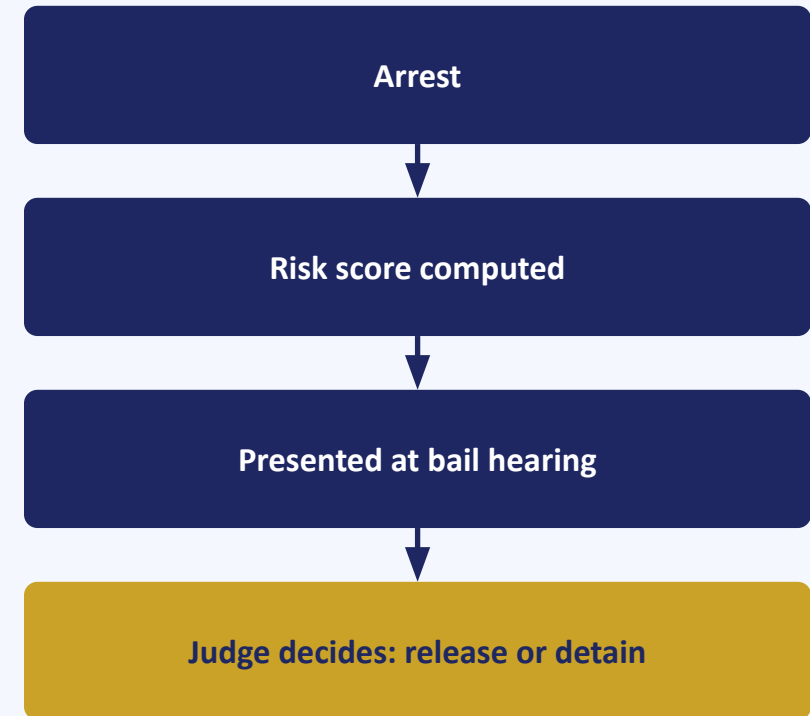
- **Output**

A continuous risk score, binarized at a decision threshold into a Low Risk / High Risk label.

- **Stakes**

Applied before conviction — it shapes bail amount and release conditions for individuals presumed innocent.

DEPLOYMENT CONTEXT



California's DTC confirmed this system in active CDCR use, June 2026 — a bill barring sole reliance on it failed to pass (Johnson, 2026).

Boundary Fragility Near the Threshold

We tested three defendants whose scores sat within 0.008 of the 0.50 decision line, then added one prior conviction to each.



What Actually Drives the Boundary?

Nearest-counterfactual test (L1 distance) on Datapoint 464 after its edit — the closest oppositely-labeled record in the dataset:

Feature	DP 464 — High Risk (0.616)	DP 2047 — Low Risk (0.453)
priors_count	2	0
age	22	22
race	Caucasian	Caucasian
sex	Male	Male
juvenile counts	identical across 3 fields	identical across 3 fields

1 of 9

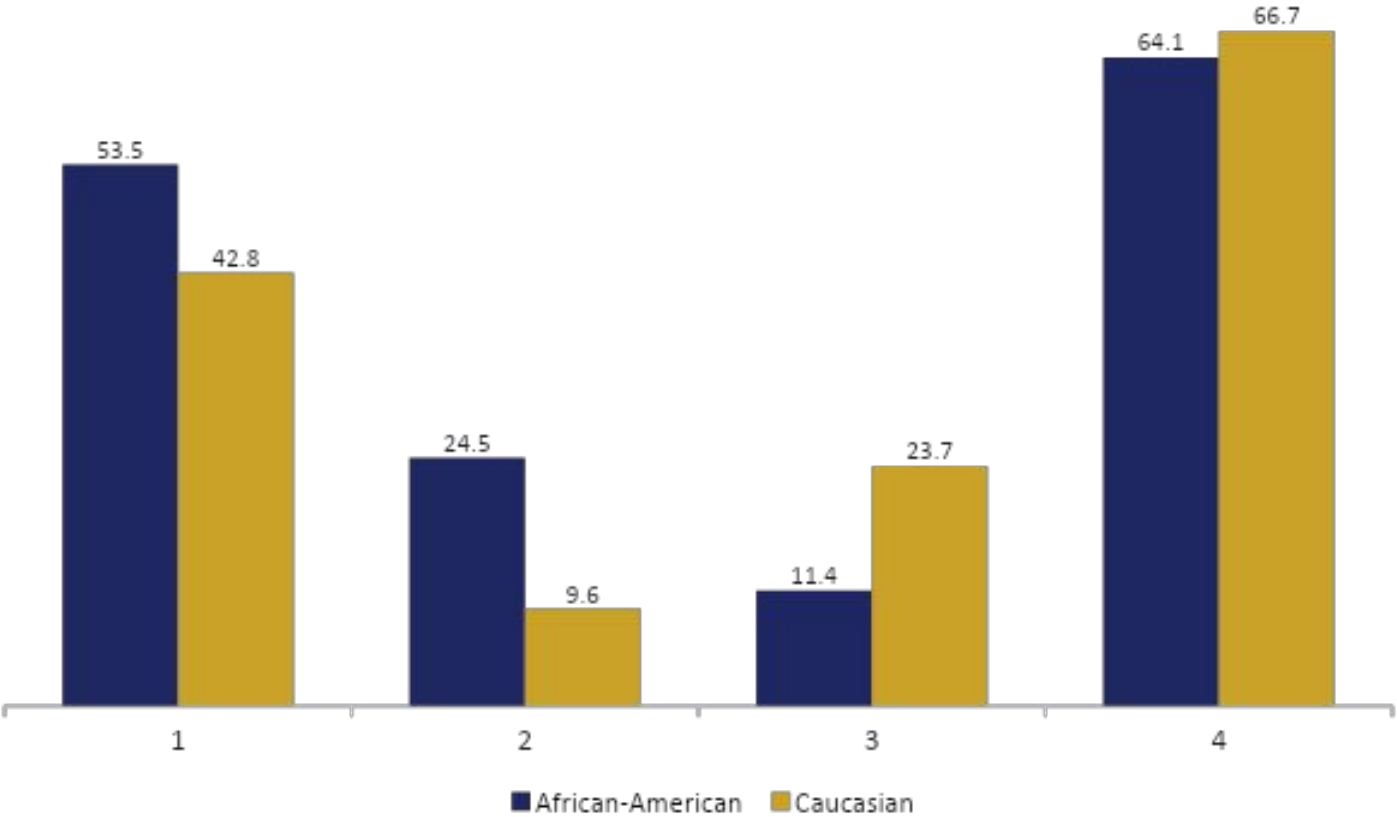
recorded features differs
between the two records

*Race, age, sex, and juvenile history are identical.
For this instance, the local boundary is governed by
criminal-history — not a demographic proxy.*

Caveat: this is an instance-specific result. It does not rule out group-level disparate impact — that's a separate question, addressed next.

Baseline Error Rates Diverge by Race

Same 0.50 threshold, both groups. n = 1,904 African-American · 1,111 Caucasian defendants.



FPR GAP

14.9 pts

African-American defendants wrongly flagged High Risk far more often

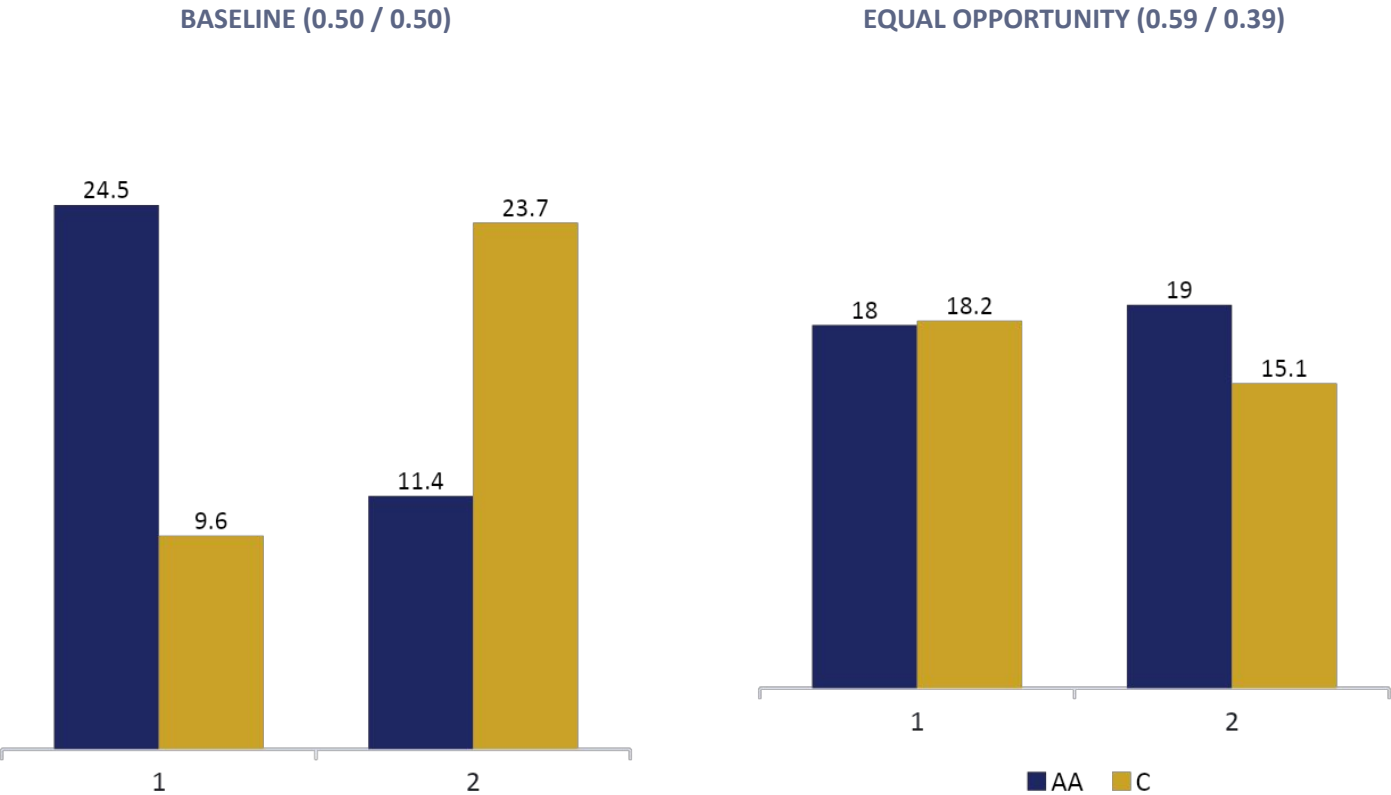
FNR GAP

12.3 pts

Caucasian defendants wrongly cleared as Low Risk far more often

Equal Opportunity Narrows the Gap

Group-specific thresholds: African-American 0.50 → 0.59 · Caucasian 0.50 → 0.39



D_FAIR

27.2

↓

4.1

85% reduction in combined FPR + FNR disparity

Thresholds applied: AA 0.59 · Caucasian 0.39

The Ethical Tax — and Which Error Costs More

ETHICAL TAX

0.63 pt

Weighted accuracy: 65.1% → 64.4%

In governance terms: a small, diffuse predictive cost purchased a large, targeted reduction in cross-racial harm.

WHICH ERROR IS MORE SEVERE?

FPR → Liberty Harm

Over-incarceration of people who would not reoffend.

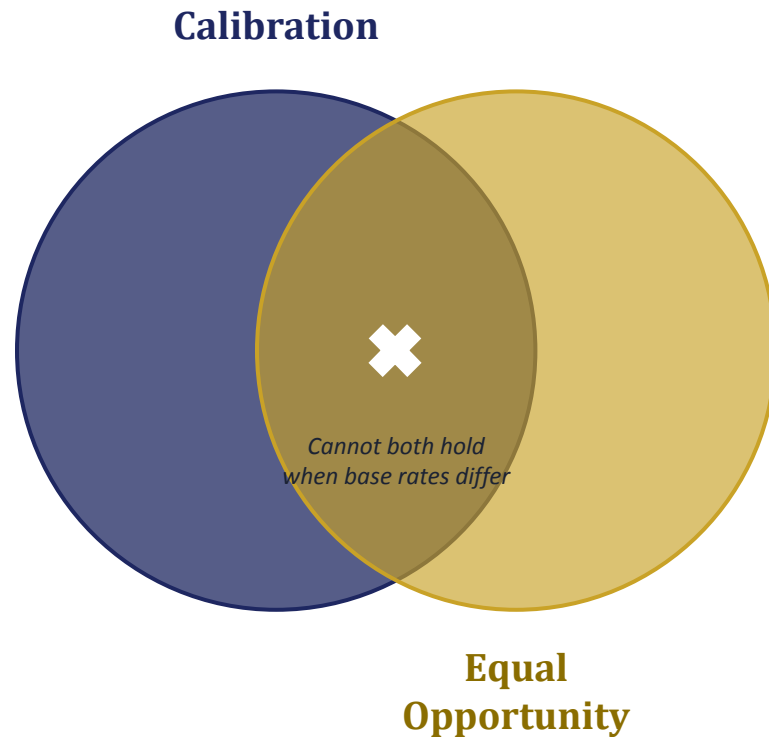
FNR → Public-Safety Harm

Releasing people who will reoffend.

Our position: liberty harm is the more severe cost in this deployment context.

- Detention here is pre-conviction — the defendant is presumed innocent.
- Wrongful detention is a certain, immediate harm; reoffense risk is probabilistic.
- The baseline FPR gap (14.9 pts) exceeded the FNR gap (12.3 pts), and fell involuntarily on one group.

Kleinberg's Impossibility Theorem



Calibration: a 70% risk score means $\approx 70\%$ actually reoffend — independent of race.

Equal Opportunity: an actual reoffender is equally likely to be flagged High Risk, regardless of race.

Kleinberg et al. (2016): when base rates differ across groups, no classifier — short of a perfect predictor — can satisfy both simultaneously.

10.7 pts

base-rate gap
(53.5% vs 42.8%)

0.71 AUC

far from a perfect
predictor

Independently confirmed in 2026: this same FPR/FNR asymmetry held across every modern classifier and debiasing technique tested — including explicit removal of the race feature (Kienzle et al., 2026).

Defending Our Fairness Weighting (λ)

$$\min_{\theta} J(\theta) + \lambda \cdot D_{\text{fair}}(P_{\theta})$$

$$J(\theta) = 1 - \text{Accuracy} \quad D_{\text{fair}}(P_{\theta}) = |FPR_{AA} - FPR_C| + |FNR_{AA} - FNR_C|$$

Our operating point — moderate λ

$\lambda = 0$
Accuracy-only

$\lambda \rightarrow \infty$
Fairness-only

$D_{\text{fair}} 27.2 \rightarrow 4.1$ (85%↓) for 0.63-pt cost

This is a governance decision, not a technical parameter.

- An 85% cut in cross-racial disparity for under one accuracy point is a favorable, defensible exchange rate.
- The harm reduced (liberty harm) is concentrated and involuntary; the cost (accuracy) is small and diffuse.
- A jurisdiction with a stronger public-safety mandate could reasonably choose a lower λ — a legitimate, different policy choice, not a technical error.
- Proof this is governance, not math: a comparable federal tool (PTRR), deployed with a specified release policy, is projected to cut detention 34.2% while narrowing the racial gap (Skeem et al., 2025).

Three NIST AI RMF Controls

1**GOVERN 1.2**

Detail model testing and validation processes

Mandatory boundary-sensitivity and counterfactual probes, plus sliced FPR/FNR as a pass/fail criterion, signed off by a validation function independent of development.

Trigger: re-validate if $D_{\text{fair}} > 10$ pts, or base-rate gap shifts > 5 pts from training.

2**GOVERN 1.2**

Establish the frequency of and detail for monitoring, auditing and review processes

Quarterly re-sampling of near-threshold cases and recomputation of sliced metrics, owned by a named auditing function with escalation to legal and compliance.

Trigger: re-audit if FPR disparity > 15 pts or FNR disparity > 12 pts.

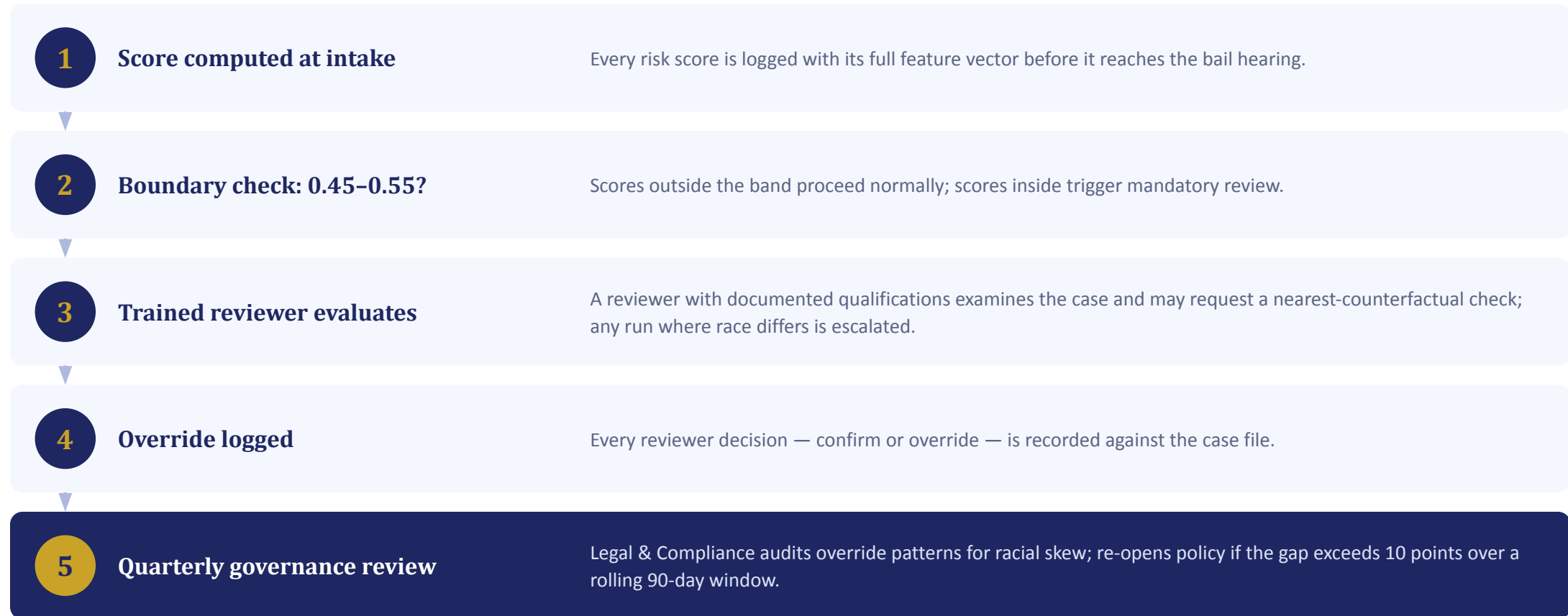
3**GOVERN 3.2**

Establish policies and procedures regarding AI actor roles, and responsibilities for human oversight of deployed systems

Mandatory human review for any score between 0.45 and 0.55, by reviewers trained on the FPR/FNR asymmetries, with every override logged.

Trigger: escalate within 0.05 of threshold; review policy if override rates differ > 10 pts by race.

Who Reviews, When, Under What Authority



What Remains True, Even After Mitigation

- **The calibration/error trade-off is mathematically fixed.**

Given the 10.7-point base-rate gap, no λ or threshold choice eliminates it — only redistributes it.

- **Equal Opportunity redistributes error; it doesn't remove it.**

African-American FNR rose from 11.4% to 19.0% — more actual reoffenders are released under the corrected model.

- **The base-rate gap likely reflects structural inequity upstream of the model.**

Differential policing and arrest patterns predate the classifier; no threshold correction repairs the underlying cause.

- **Threshold-based fixes can decay.**

Both fairness gains are sensitive to distributional shift over time — this is exactly what the audit cadence is for.

- **Human oversight cannot resolve the statistical constraint itself.**

It can catch mechanical reproduction of inequity in individual cases, but the group-level trade-off remains.

Post-Deployment Monitoring Triggers

Metric	Trigger Threshold	Action	Cadence
D_fair (combined disparity)	> 10 pts	Re-validation	Continuous
Base-rate gap shift	> 5 pts vs. training	Re-validation	Continuous
FPR disparity	> 15 pts	Re-audit	Quarterly checkpoint
FNR disparity	> 12 pts	Re-audit	Quarterly checkpoint
Score vs. active threshold	within 0.05	Mandatory human escalation	Real-time, every case
Override-rate racial gap	> 10 pts	Policy review	Rolling 90-day window

Every trigger above is active from day one of deployment and reviewed at the quarterly governance meeting alongside the sliced-metrics disclosure.

Conditions for Go-Live

1

Deploy at Equal Opportunity thresholds (0.59 African-American / 0.39 Caucasian) — not the 0.50 baseline.

2

All three NIST AI RMF controls operational before go-live: validation gate, audit cadence, human-oversight policy.

3

Mandatory human review and logging for every score in the 0.45–0.55 band.

4

Quarterly sliced-metrics disclosure to the compliance/governance board.

5

Automatic suspension and re-validation the moment any monitoring trigger fires.

6

Documented acknowledgment that the base-rate gap reflects upstream structural factors — never grounds for race-based manual overrides.

CLOSING

Final Recommendation

CONDITIONAL GO

85%

↓ in cross-group
error disparity

0.63 pt

accuracy cost of
the correction

3

NIST AI RMF
controls required

0.05

escalation band around
the active threshold

The disparities uncovered here are structural, not a tuning failure — and the same statistical constraint can be disparity-reducing or disparity-reproducing, depending on governance alone.

With this system now confirmed in active use across California, Equal Opportunity thresholds, three NIST-aligned controls, and a defined monitoring plan are what put us on the disparity-reducing side of that choice.